This week was good. I was able to make some good progress. I made my own branch and added a folder where I put my code. In this folder, I have written code that makes all of the models included in the tensorflow tutorial with the data that you mentioned in the previous meeting, from station 1 in Revised_Final_data folder. Additionally it also is tested on the ARIMA model. Then to see which features were the most important, I included shapley values and found solar radiation to be the most important followed by the temperature of the ground. I then tested the models without these important values and screenshotted some results.

Before 9/25 meeting

My work -

Summary on the first station's data

```
Model Performance Summary:
Model: Baseline Model - Loss: 0.0002, MAE: 0.0037
Model: Linear Model - Loss: 0.0032, MAE: 0.0289
Model: Dense Model - Loss: 0.0030, MAE: 0.0281
Model: CNN Model - Loss: 0.0001, MAE: 0.0031
Model: RNN Model - Loss: 0.0001, MAE: 0.0056
Model: Autoregressive Model - Loss: 0.0032, MAE: 0.0231
Model: ARIMA Model - MSE: 0.0430, RMSE: 0.2074, MAE: 0.1649, R²: -0.4125
zanderh@coral:~/tx-soil-moisture/code$
```

Shapley values- seeing which features are most important to the model

```
You have provided over 5k background samples! For better perform
`tf.keras.backend.set_learning_phase` is deprecated and will be
ent of the `__call__` method of your layer or model.

Feature Importance based on SHAP values:
Srad: 0.0034
T_5: 0.0027
T_20: 0.0024
T_50: 0.0021
T_10: 0.0020
RH: 0.0011
Windspeed: 0.0006
Tair: 0.0006
Winddirection: 0.0004
Ppt: 0.0003

Model Performance Summary:
Model: Dense Model - Loss: 0.0010, MAE: 0.0257
zanderh@coral:~/tx-soil-moisture/code$
```

Without Srad, this is what the new performance summary looks like

```
Model Performance Summary:
Model: Baseline Model - Loss: 0.1290, MAE: 0.2879
Model: Linear Model - Loss: 0.0018, MAE: 0.0345
Model: Dense Model - Loss: 0.0010, MAE: 0.0253
Model: CNN Model - Loss: 0.0011, MAE: 0.0259
Model: RNN Model - Loss: 0.0011, MAE: 0.0248
Model: Autoregressive Model - Loss: 0.0014, MAE: 0.0275
Model: ARIMA Model - MSE: 0.0040, RMSE: 0.0631, MAE: 0.0502, R²: -0.4125
zanderh@coral:~/tx-soil-moisture/code$
```

Features - everything except T_5

```
329/329 [==============================] - 1s 2ms/step - loss: 0.0014 - mae:
test loss: 0.001411012257449329, test mae: 0.02882879227399826


Model Performance Summary:
Model: Baseline Model - Loss: 0.1290, MAE: 0.2879
Model: Linear Model - Loss: 0.0018, MAE: 0.0346
Model: Dense Model - Loss: 0.0012, MAE: 0.0280
Model: CNN Model - Loss: 0.0011, MAE: 0.0272
Model: RNN Model - Loss: 0.0012, MAE: 0.0271
Model: Autoregressive Model - Loss: 0.0014, MAE: 0.0288
Model: ARIMA Model - MSE: 0.0040, RMSE: 0.0631, MAE: 0.0502, R²: -0.4125
```

10/2 MEETING

MSE on each model before removing any of the features. parameter is always

```
# parameters - more generalized
TARGET_COL = "SWC_5"
TRAIN_SPLIT = 0.7
VAL_SPLIT = 0.2
WINDOW_SIZE = 24 * 7
SHIFT_AMT = 10
PAT = 3
MAX_EPOCHS = 25
BATCH_SIZE = 128
```

```
BiLSTM Model MSE: 0.005176883190870285
```

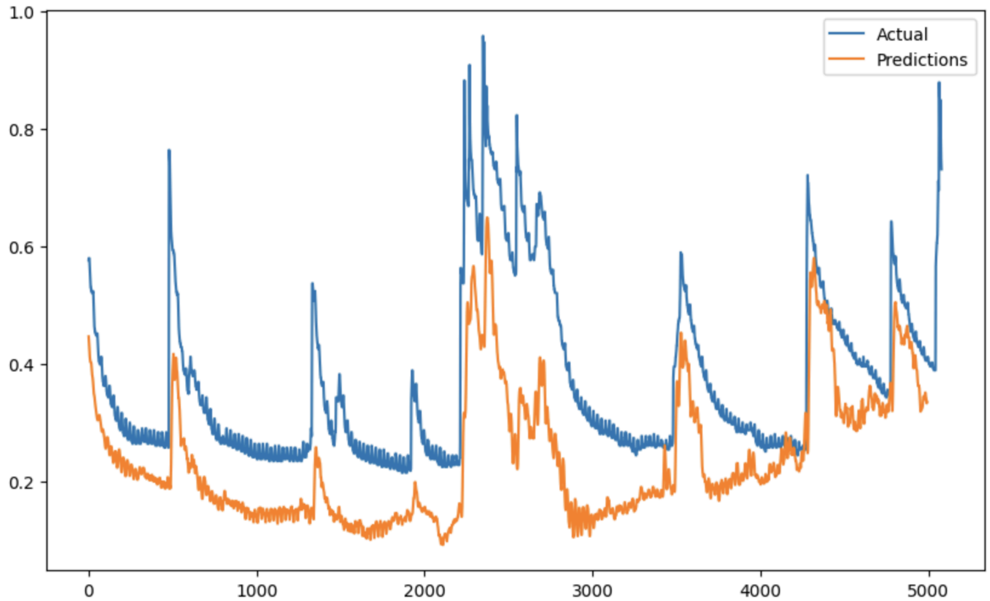
```
315/315 [==============================] - 1s 3ms/s
absolute_error: 2.9348 - val_mean_squared_error: 13
ARIMA Model MSE: 0.003755368085690797
Linear Model MSE: 102.67269897460938
Autoregressive Model MSE: 99.14004516601562
dense Model MSE: 1.4055372476577759
ARIMA Model MSE: 0.003755368085690797
RNN Model MSE: 0.0019769289065152407
CNN Model MSE: 13.255877494812012
zanderh@coral:~/tx-soil-moisture/Alex$ 
```

Graphs -

biLSTM -

```
[37] plot_single_pred(bi_lstm_model,'BiLSTM', test_dataset, test_steps, y_test, batch_size=BATCH_SIZE)
```

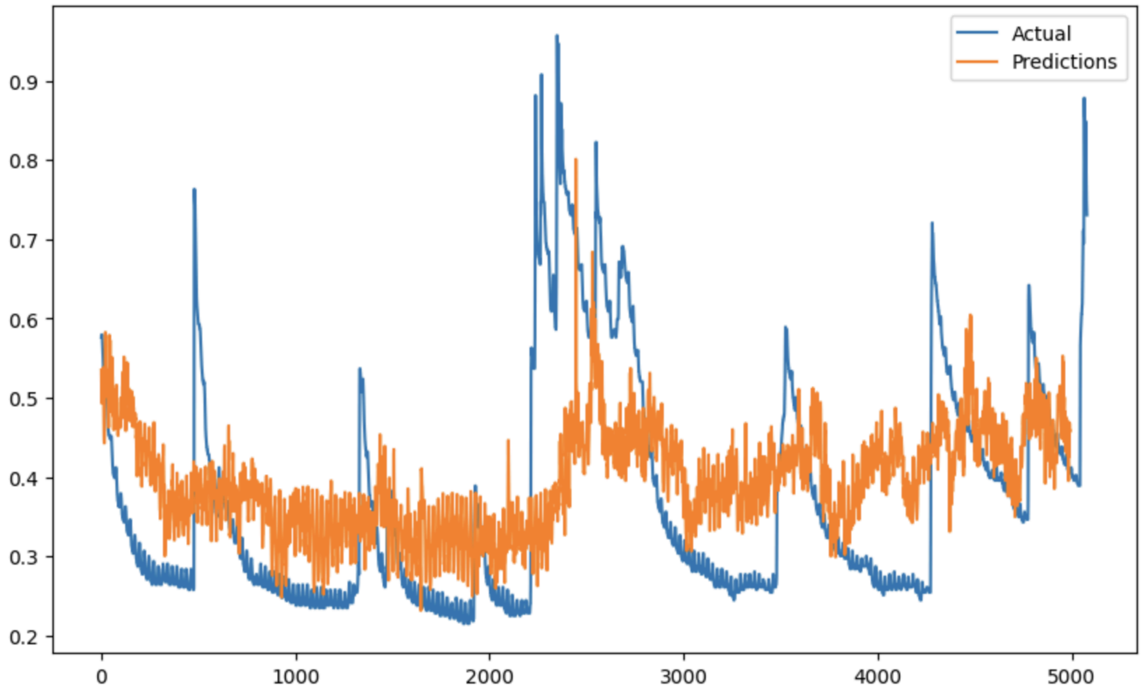
```
39/39 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step
{'Predictions': array([0.4463259 , 0.44244823, 0.43767148, ..., 0.33642453, 0.33299166,
       0.3349844 ], dtype=float32),
 'Actual': array([0.57619943, 0.57948471, 0.57948471, ..., 0.74703377, 0.73717794,
       0.73060739])}
```



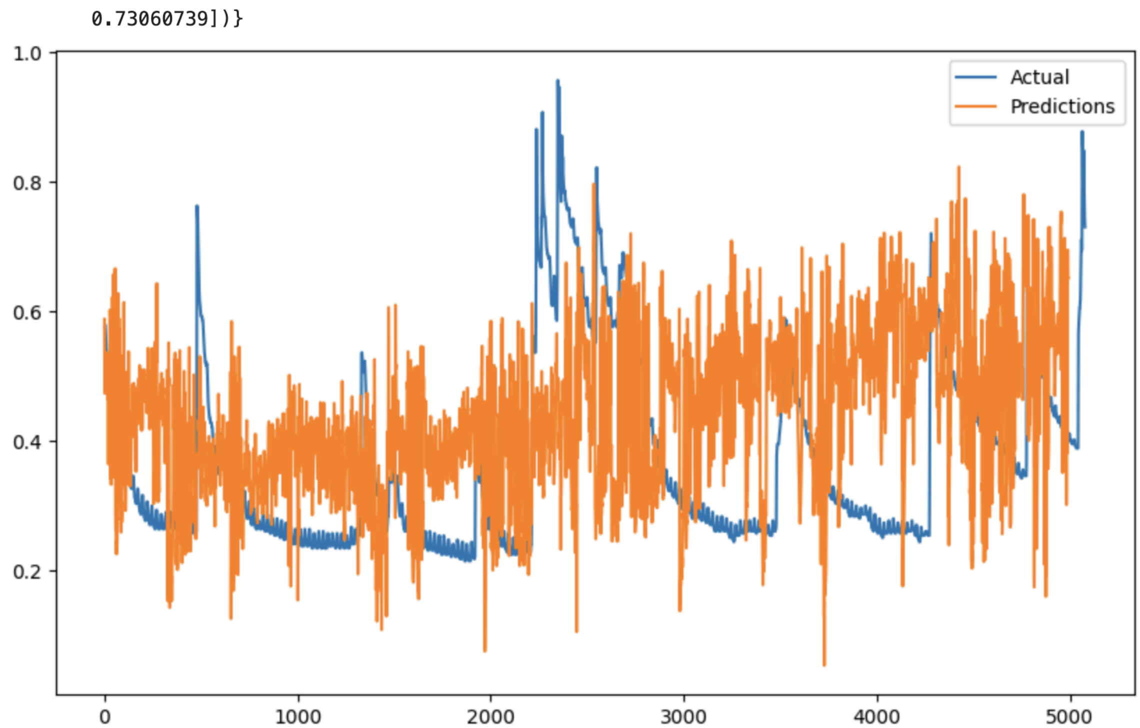
linear

```
plot_single_pred(linear_model, 'linear', test_dataset, test_steps, y_test, batch_size=BATCH_SIZE)
```

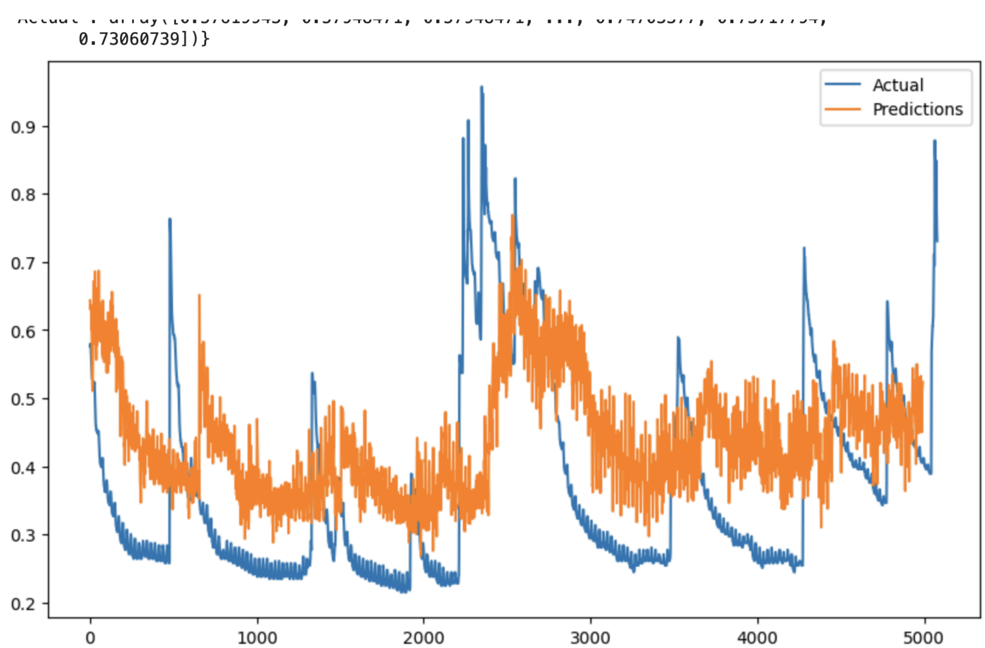
```
39/39 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
asd
{'Predictions': array([0.53510195, 0.5070601 , 0.49300486, ..., 0.46729344, 0.45674986,
        0.458418  ], dtype=float32),
 'Actual': array([0.57619943, 0.57948471, 0.57948471, ..., 0.74703377, 0.73717794,
        0.73060739])}
```



Autoregressive -

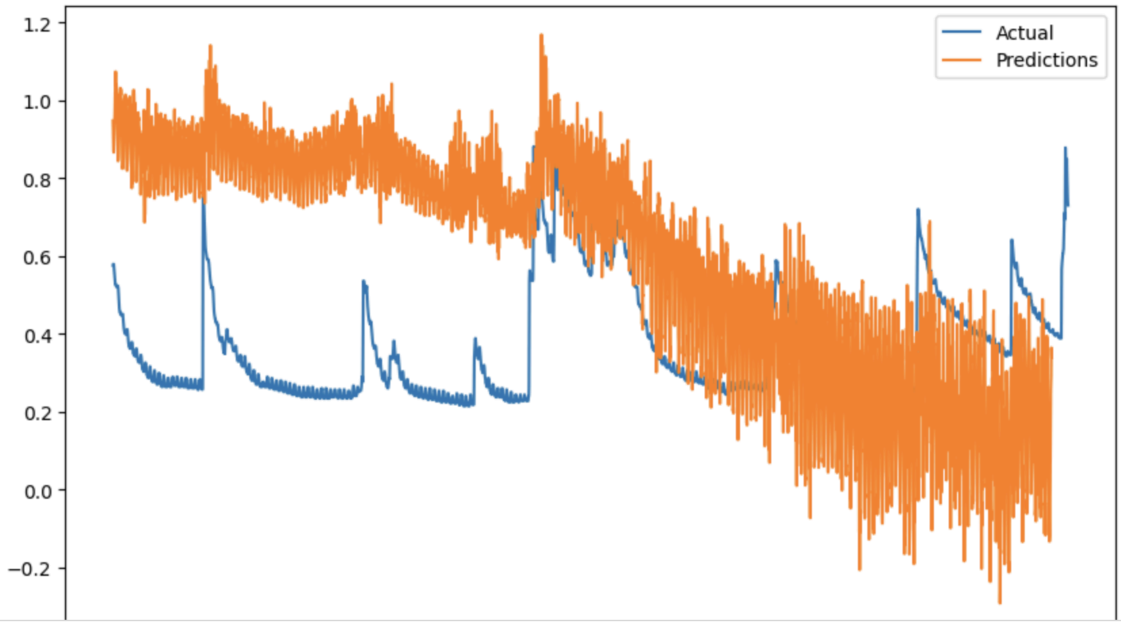

0.73060739])}
Actual
Predictions

Dense -


0.73060739])}
Actual
Predictions

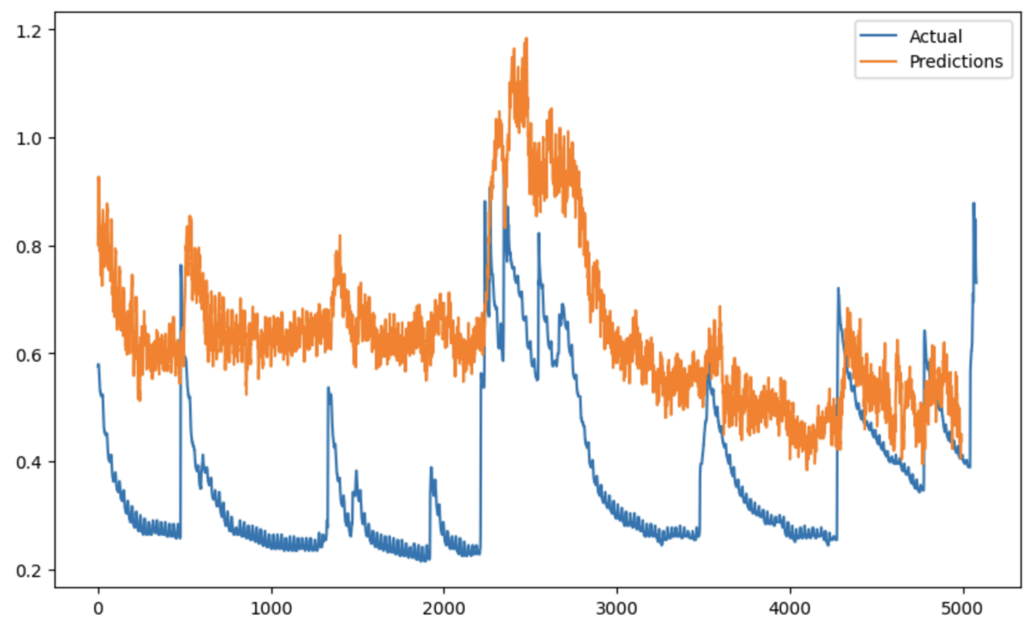
RNN

```
0.73060739])}
```

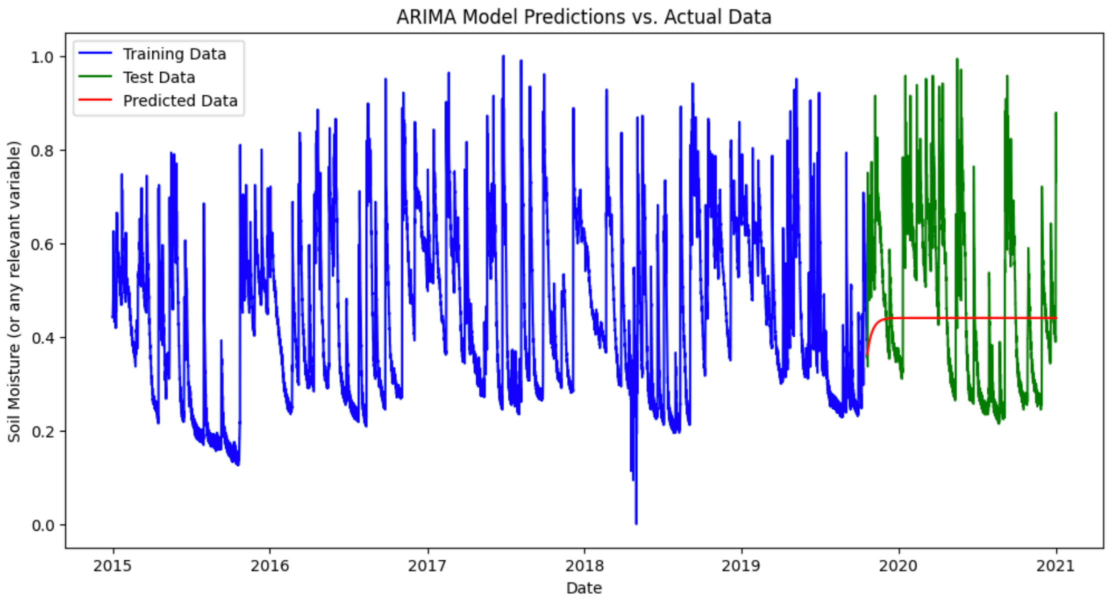


CNN

```
'Actual': array([0.57619943, 0.57948471, 0.57948471, ..., 0.74703377, 0.73717794,
       0.73060739])}
```



ARIMA (simple) with (1,0,0) for p,d,q



Now for most important features of the best model - Dense and RNN. I would get rid of one feature at a time for Dense and RNN since they performed well.

Dense

```
mse results after removing each feature:
feature: Ppt, mse: 0.2973490059375763
feature: SWC_10, mse: 0.4120842218399048
feature: SWC_20, mse: 0.31319689750671387
feature: SWC_50, mse: 0.20060929656028748
feature: T_5, mse: 0.1461014300584793
feature: T_10, mse: 0.1024278849363327
feature: T_20, mse: 0.1618952453136444
feature: T_50, mse: 0.024383798241615295
feature: Ppt.1, mse: 0.08615513890981674
feature: Tair, mse: 0.1386115700006485
feature: RH, mse: 0.02551613561809063
feature: Windspeed, mse: 0.02824592962861061
feature: Winddirection, mse: 0.027985772117972374
feature: Srad, mse: 0.0490608848631382
feature: Latitude, mse: 0.06385380774736404
feature: Longitude, mse: 0.0279052946716547

sorted mse (higher mse indicates more important feature):
feature: SWC_10, mse: 0.4120842218399048
feature: SWC_20, mse: 0.31319689750671387
feature: Ppt, mse: 0.2973490059375763
feature: SWC_50, mse: 0.20060929656028748
feature: T_20, mse: 0.1618952453136444
feature: T_5, mse: 0.1461014300584793
feature: Tair, mse: 0.1386115700006485
feature: T_10, mse: 0.1024278849363327
feature: Ppt.1, mse: 0.08615513890981674
feature: Latitude, mse: 0.06385380774736404
feature: Srad, mse: 0.0490608848631382
feature: Windspeed, mse: 0.02824592962861061
feature: Winddirection, mse: 0.027985772117972374
feature: Longitude, mse: 0.0279052946716547
feature: RH, mse: 0.02551613561809063
feature: T_50, mse: 0.024383798241615295
```

RNN

```
mse results after removing each feature for RNN:
feature: Ppt, mse: 0.002112413989380002
feature: SWC_10, mse: 0.00177768396679312
feature: SWC_20, mse: 0.0016875251894816756
feature: SWC_50, mse: 0.0017219082219526172
feature: T_5, mse: 0.0016869427636265755
feature: T_10, mse: 0.0016886289231479168
feature: T_20, mse: 0.0015538223087787628
feature: T_50, mse: 0.0015288705471903086
feature: Ppt.1, mse: 0.0017883736873045564
feature: Tair, mse: 0.0016512329457327724
feature: RH, mse: 0.0018760806415230036
feature: Windspeed, mse: 0.0014988677576184273
feature: Winddirection, mse: 0.001475014491006732
feature: Srad, mse: 0.001578167430125177
feature: Latitude, mse: 0.0015251750592142344
feature: Longitude, mse: 0.0017474588239565492

sorted mse (higher mse indicates more important feature in RNN):
feature: Ppt, mse: 0.002112413989380002
feature: RH, mse: 0.0018760806415230036
feature: Ppt.1, mse: 0.0017883736873045564
feature: SWC_10, mse: 0.00177768396679312
feature: Longitude, mse: 0.0017474588239565492
feature: SWC_50, mse: 0.0017219082219526172
feature: T_10, mse: 0.0016886289231479168
feature: SWC_20, mse: 0.0016875251894816756
feature: T_5, mse: 0.0016869427636265755
feature: Tair, mse: 0.0016512329457327724
feature: Srad, mse: 0.001578167430125177
feature: T_20, mse: 0.0015538223087787628
feature: T_50, mse: 0.0015288705471903086
feature: Latitude, mse: 0.0015251750592142344
feature: Windspeed, mse: 0.0014988677576184273
feature: Winddirection, mse: 0.001475014491006732
```

Now just have one feature in the model at a time with the label as SWC_5. Which feature is most important?

Dense model

```
0024 - val_mean_absolute_error: 0.0425 - val_mean_squared_error: 0.0024

mse results after adding each feature for Dense:
feature: Ppt, mse: 0.0013432246632874012
feature: SWC_10, mse: 0.0013630182947963476
feature: SWC_20, mse: 0.0013535980833694339
feature: SWC_50, mse: 0.0013718537520617247
feature: T_5, mse: 0.0016843334306031466
feature: T_10, mse: 0.002492288127541542
feature: T_20, mse: 0.0025393993128091097
feature: T_50, mse: 0.003875530092045665
feature: Ppt.1, mse: 0.001358287176117301
feature: Tair, mse: 0.0020136800594627857
feature: RH, mse: 0.001957417232915759
feature: Windspeed, mse: 0.0013964513782411814
feature: Winddirection, mse: 0.0032401778735220432
feature: Srad, mse: 0.003487402107566595
feature: Latitude, mse: 0.005155886523425579
feature: Longitude, mse: 0.002358902944251895

sorted mse (lower mse indicates more important feature in Dense):
feature: Ppt, mse: 0.0013432246632874012
feature: SWC_20, mse: 0.0013535980833694339
feature: Ppt.1, mse: 0.001358287176117301
feature: SWC_10, mse: 0.0013630182947963476
feature: SWC_50, mse: 0.0013718537520617247
feature: Windspeed, mse: 0.0013964513782411814
feature: T_5, mse: 0.0016843334306031466
feature: RH, mse: 0.001957417232915759
feature: Tair, mse: 0.0020136800594627857
feature: Longitude, mse: 0.002358902944251895
feature: T_10, mse: 0.002492288127541542
feature: T_20, mse: 0.0025393993128091097
feature: Winddirection, mse: 0.0032401778735220432
feature: Srad, mse: 0.003487402107566595
feature: T_50, mse: 0.003875530092045665
feature: Latitude, mse: 0.005155886523425579
zanderh@coral:~/tx-soil-moisture/Alex$
```

Report - Have a much cleaner version of the code that is working on my end and have generalized it for the parameters. Added all the models from tutorials plus biLSTM. I also included a very simple ARIMA model that is also working. Besides that, I have also found the most important features in the RNN and Dense models (the best performing models according to MSE). The most important features for label - SWC_5 are the SWC_10, SWC_15, SWC_20. This makes sense because they are the label but at different depth. Also PPT(precipitation is closely related) which makes sense. Least related features have to do with wind and temperature. Also, seems like dense model has most accurate predictions that make sense in comparison to RNN.

Report on 10/7

Got rid of other SWC values when label is SWC_5, included MAE and MAPE, generalized the code to make it look more like in the /code folder.

ALL models (MSE, MAE, MAPE). How did they perform?

```
BiLSTM Model MSE: {'MSE': 0.001740805571898818, 'MAE': 0.0345991775393486, 'MAPE': 28.929332733154297}
Linear Model MSE: {'MSE': 0.24864891171455383, 'MAE': 0.4092881679534912, 'MAPE': 297.9903259277344}
Autoregressive Model MSE: {'MSE': 0.5369340777397156, 'MAE': 0.596554696559906, 'MAPE': 429.9562072753906}
Dense Model MSE: {'MSE': 0.8959733247756958, 'MAE': 0.7312438488006592, 'MAPE': 526.0953979492188}
RNN Model MSE: {'MSE': 0.01126321405172348, 'MAE': 0.09171751886606216, 'MAPE': 59.21674346923828}
CNN Model MSE: {'MSE': 0.0028233621269464493, 'MAE': 0.047064099460840225, 'MAPE': 38.610843658447266}
zanderh@coral:~/tx-soil-moisture/Alex$
```

Getting rid of values

Dense model

```
MSE results after removing each feature for Dense: (higher indicates more important)
Feature: Ppt, MSE: 0.04722440242767334
Feature: T_10, MSE: 0.034118134528398514
Feature: T_50, MSE: 0.033186789602041245
Feature: T_5, MSE: 0.03267695754766464
Feature: Windspeed, MSE: 0.03156442195177078
Feature: Ppt.1, MSE: 0.027838028967380524
Feature: T_20, MSE: 0.025254007428884506
Feature: Tair, MSE: 0.020198963582515717
Feature: RH, MSE: 0.019290966913104057
Feature: Winddirection, MSE: 0.014300394803285599
Feature: Latitude, MSE: 0.00290765636600554
Feature: Srad, MSE: 0.002877915743738413
Feature: Longitude, MSE: 0.0025543200317770243

MAE results after removing each feature for Dense:
Feature: Ppt, MAE: 0.15536941587924957
Feature: T_10, MAE: 0.1552729308605194
Feature: T_5, MAE: 0.13955800235271454
Feature: T_50, MAE: 0.13636744022369385
Feature: Ppt.1, MAE: 0.13588963449001312
Feature: T_20, MAE: 0.12431114166975021
Feature: Windspeed, MAE: 0.12174468487501144
Feature: Tair, MAE: 0.10297933220863342
Feature: RH, MAE: 0.10147034376859665
Feature: Winddirection, MAE: 0.08722971379756927
Feature: Srad, MAE: 0.047448884695768356
Feature: Latitude, MAE: 0.04492054134607315
Feature: Longitude, MAE: 0.04316248372197151

MAPE results after removing each feature for Dense:
Feature: T_10, MAPE: 138.9252471923828
Feature: T_50, MAPE: 125.18655395507812
Feature: Ppt, MAPE: 111.64459228515625
Feature: T_5, MAPE: 109.33294677734375
Feature: T_20, MAPE: 98.8306655883789
Feature: Ppt.1, MAPE: 93.34263610839844
Feature: Windspeed, MAPE: 83.801513671875
Feature: Tair, MAPE: 68.762451171875
Feature: RH, MAPE: 65.95391845703125
Feature: Winddirection, MAPE: 60.94920349121094
Feature: Srad, MAPE: 40.56770706176758
Feature: Latitude, MAPE: 34.57167053222656
Feature: Longitude, MAPE: 33.223209381103516
```

RNN model took too long - process was killed by UTCS machines

Adding one feature at a time
Now just have one feature in the model at a time with the label as SWC_5. Which feature is most important?

Dense model

```
MSE results after adding each feature for Dense: (lower indicates more important)
Feature: Windspeed, MSE: 0.001453785109333694
Feature: Ppt.1, MSE: 0.0014605943579226732
Feature: Tair, MSE: 0.001941379625350237
Feature: Srad, MSE: 0.0020582028664648533
Feature: Longitude, MSE: 0.0022095385938882828
Feature: Latitude, MSE: 0.0025860636960715055
Feature: Ppt, MSE: 0.0025896290317177773
Feature: Winddirection, MSE: 0.0026602023281157017
Feature: RH, MSE: 0.0026813114527612925
Feature: T_10, MSE: 0.0029073311015963554
Feature: T_20, MSE: 0.006790510844439268
Feature: T_5, MSE: 0.0076898736879229546
Feature: T_50, MSE: 0.015561138279736042

MAE results after adding each feature for Dense:
Feature: Windspeed, MAE: 0.02978871762752533
Feature: Ppt.1, MAE: 0.030241044238209724
Feature: Tair, MAE: 0.033496223390102386
Feature: Ppt, MAE: 0.034729305654764175
Feature: Srad, MAE: 0.03823687508702278
Feature: Longitude, MAE: 0.04095481336116791
Feature: Latitude, MAE: 0.042260490357875824
Feature: Winddirection, MAE: 0.04464005306363106
Feature: RH, MAE: 0.04488897696137428
Feature: T_10, MAE: 0.04632143676280975
Feature: T_20, MAE: 0.06805134564638138
Feature: T_5, MAE: 0.0767153650522232
Feature: T_50, MAE: 0.1094316840171814

MAPE results after adding each feature for Dense:
Feature: Windspeed, MAPE: 21.643840789794922
Feature: Tair, MAPE: 21.651439666748047
Feature: Ppt, MAPE: 21.98928451538086
Feature: Ppt.1, MAPE: 22.288612365722656
Feature: Latitude, MAPE: 29.596118927001953
Feature: Srad, MAPE: 30.319896697998047
Feature: RH, MAPE: 34.28999710083008
Feature: Winddirection, MAPE: 34.349586486816406
Feature: Longitude, MAPE: 36.10234451293945
Feature: T_10, MAPE: 41.3229866027832
Feature: T_20, MAPE: 41.36985397338867
Feature: T_5, MAPE: 51.39188003540039
Feature: T_50, MAPE: 70.10079193115234
```

RNN model took too long - process was killed by UTCS machines

Report - not too much difference from the report earlier. Ppt being so related SWC_5 makes a lot of sense. My earlier observations are pretty much the same as they are before based on this new data (with MSE, MAE, MAPE)

Will be working with Michael this week to combine our code. We already have an idea of what is different between our code and how to make it more similar

- Finding a common implementation for the windowgenerator
- Testing different input sizes and output sizes for the windows